Random-covariance Skorohod completion, trace-corrected diagonal

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claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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Purpose and scope

The random-covariance second jet and the diagonal interpolation are now exact, and the first stationary scalar cross-mode model has a strict square-first variance margin. The production lower form is not closed.

For a symmetric current-root-dependent covariance $W(\xi)$, the complete second jet is one double Gaussian divergence $\delta_\xi^2 W$. Its Young-optimized abstract quadratic/sixth-moment form cost is controlled by $K_W = \mathbb{E}[(\delta_\xi^2 W)^2 \mid \mathcal{F}_{\mathrm{past}}]$, not by the pathwise trace, operator norm, or Hilbert–Schmidt norm of W. A uniformly positive scalar covariance and a rank-one rotating projector prove that this distinction is load-bearing. Identification of these abstract moments with the production source and terminal-sextic owners remains open.

The diagonal current $b + A\xi + R(\xi)$ admits an exact trace-corrected interpolation to its decoupled endpoint. Its derivative is one tilted cross-score containing the differentiated trace forest. A scalar Hermite fixture proves that a universal nonlinear square-first normalizer is false unless its negative mean debt is first cancelled or explicitly paid.

For the smallest stationary scalar two-frequency $k{:}2k$ whole-output model, the exact covariance S_X, packet P, variance, third moment, and $H = \mathbb{E}\|S_X\|^2_{\mathfrak{S}_2}$ are computed. The identity $H - 2\,\mathrm{Var}(P) = (v^4 + 16v^2w^2 + 16w^4)/4$ gives strict small-q room when $v + w > 0$. A sharp pointwise payment proves a related normalizer, but that payment reintroduces the nonsummable $A^2|k|^{-2}$ baseline. The bare all-q complete-cluster inequality remains a live, unproved target.

No statement below proves the adapted production cluster, the one-use source-action lower bound, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson, any removal, an interacting measure, or Sector-A closure. A13 remains T4.

The scope is finite cutoff and fixed positive density floor. The random-W and owner questions use the A1/R-063/R-107 production conventions, while Section 5 is a stationary scalar two-frequency model and is not asserted to be an embedded full A1 production cluster. All conditional differentiations below are cylindrical and justified first for smooth data. The random-W theorem assumes W lies in the conditional domain of δ^2, the displayed products are integrable, and Gaussian boundary terms vanish. Its Sobolev corollary uses the full tensor norms of DW and D^2W, including both matrix and Malliavin indices.

1 Complete random-W score transfer

Condition on $\mathcal{F}_{\mathrm{past}}$. Let $\xi \sim N(0, \mathrm{I}_d)$, let H be a real Hilbert space, let $h : \mathbb{R}^d \to H$, and let $W : \mathbb{R}^d \to \mathrm{Sym}_d$ be positive semidefinite. Put

$$(\operatorname{div} W)_i = \sum_j \partial_j W_{ij}, \qquad \operatorname{div}^2 W = \sum_{i,j} \partial_{ij} W_{ij}. \tag{1.1}$$

The current-root Gaussian divergence and its second iterate are

$$(\delta_\xi W)_i = (W\xi - \operatorname{div} W)_i, \tag{1.2}$$

$$\boxed{\delta_\xi^2 W = \xi^T W\xi - \operatorname{Tr} W - 2\xi \cdot \operatorname{div} W + \operatorname{div}^2 W.} \tag{1.3}$$

Define the complete signed second jet

$$\mathfrak{F}_W(h) = \mathbb{E}_\xi \left[\|D_\xi h\, W^{1/2}\|_{\mathfrak{S}_2}^2 + \langle h, \operatorname{Tr}_{\mathbb{R}^d}(W D_\xi^2 h)\rangle \mid \mathcal{F}_{\text{past}} \right]. \tag{1.4}$$

Theorem 1.1 (random-covariance double-divergence transfer)

Under the preceding domain and integrability assumptions,

$$\boxed{\mathfrak{F}_W(h) = \frac{1}{2}\mathbb{E}_\xi[\|h\|^2 \delta_\xi^2 W \mid \mathcal{F}_{\text{past}}].} \tag{1.5}$$

Equivalently,

$$\boxed{\mathfrak{F}_W(h) = \mathbb{E}_\xi[\langle h, D_\xi h(W\xi - \operatorname{div} W)\rangle \mid \mathcal{F}_{\text{past}}].} \tag{1.6}$$

Proof. For $g = \|h\|^2$,

$$\frac{1}{2}\partial_{ij} g = \langle \partial_i h, \partial_j h\rangle + \langle h, \partial_{ij} h\rangle. \tag{1.7}$$

Contract (1.7) with W_{ij} and use twice the adjoint relation between D_ξ and δ_ξ. This proves (1.5). Applying the adjoint relation once proves (1.6). ☐

The naive frozen-coefficient comparator is

$$\frac{1}{2}\mathbb{E}_\xi[\|h\|^2(\xi^T W\xi - \operatorname{Tr} W) \mid \mathcal{F}_{\text{past}}]. \tag{1.8}$$

It is not obtained by actually freezing a random W. The exact correction between (1.5) and (1.8) is

$$\boxed{\mathcal{R}_W(h) = \mathbb{E}_\xi \left[\|h\|^2 \left\{ -\xi \cdot \operatorname{div} W + \frac{1}{2}\operatorname{div}^2 W \right\} \mid \mathcal{F}_{\text{past}} \right].} \tag{1.9}$$

Thus the DW, D^2W forest is not deleted; it is assembled into one raw double-divergence coordinate.

Theorem 1.2 (Young-optimized conditional form cost)

Let

$$K_W = \mathbb{E}_\xi[(\delta_\xi^2 W)^2 \mid \mathcal{F}_{\text{past}}], \quad X = \mathbb{E}_\xi[\|h\|^2 \mid \mathcal{F}_{\text{past}}], \quad Y = \mathbb{E}_\xi[\|h\|^6 \mid \mathcal{F}_{\text{past}}]. \tag{1.10}$$

Then

$$\boxed{|\mathfrak{F}_W(h)| \le \frac{1}{2}K_W^{1/2} X^{1/4} Y^{1/4}.} \tag{1.11}$$

For every $q, \eta, \zeta > 0$,

$$\boxed{-\frac{q}{2}\mathfrak{F}_W(h) \ge -\eta X - \zeta Y - \frac{q^2 K_W}{128\sqrt{\eta\zeta}}.} \tag{1.12}$$

Proof. Equation (1.5), Cauchy–Schwarz, and $\mathbb{E}\|h\|^4 \le X^{1/2} Y^{1/2}$ give (1.11). Apply the optimized scalar inequality

$$A x^{1/4} y^{1/4} \le \eta x + \zeta y + \frac{A^2}{8\sqrt{\eta\zeta}} \tag{1.13}$$

with $A = qK_W^{1/2}/4$. □

If W is strict-past measurable, then

$$K_W = 2\|W\|^2_{\mathfrak{S}_2}. \tag{1.14}$$

At $q = 10/9$, the random-score coefficient in (1.12) is 25/2592; (1.14) recovers the R-109 coefficient 25/1296 exactly.

Corollary 1.3 (dimension-free chaos–Sobolev sufficient cost)

Write the conditional chaos expansion

$$W = \sum_{n\geq 0} I_n(f_n), \qquad f_n \in (\mathbb{R}^d)^{\odot n} \otimes (\mathbb{R}^d)^{\odot 2}. \tag{1.15}$$

Full symmetrisation of the $n+2$ indices does not increase norm. Since

$$(n+1)(n+2) = n(n-1) + 4n + 2, \tag{1.16}$$

one obtains

$$\boxed{K_W \leq \mathbb{E}_\xi[\|D^2W\|^2 + 4\|DW\|^2 + 2\|W\|^2 \mid \mathcal{F}_{\mathrm{past}}].} \tag{1.17}$$

The norms in (1.17) are full tensor Hilbert norms. Equality holds in one dimension and whenever every chaos kernel is symmetric across all slots.

This is a sufficient production gate, not an accepted production estimate. An arbitrary Boue–Dupuis control has only the declared source energy; no uniform DW, D^2W budget may be assumed.

2 Static covariance norms do not control current-root rotation

Proposition 2.1 (uniformly positive scalar no-go)

Let $G \sim N(0,1)$, $0 < \epsilon < 1$, $\tau \in \{+1, -1\}$, and

$$h_M(G) = M^{-1/2}\sin(MG), \qquad w_M^\tau(G) = 1 + \tau\epsilon\cos(2MG). \tag{2.1}$$

Then $1 - \epsilon \leq w_M^\tau \leq 1 + \epsilon$, but

$$\boxed{\mathfrak{F}_{w_M^\tau}(h_M) = M\left[e^{-2M^2} + \frac{\tau\epsilon}{2}(1 + e^{-8M^2})\right] \sim \frac{\tau\epsilon}{2}M.} \tag{2.2}$$

Meanwhile

$$X_M = \frac{1 - e^{-2M^2}}{2M} \longrightarrow 0, \tag{2.3}$$

$$Y_M = \frac{10 - 15e^{-2M^2} + 6e^{-8M^2} - e^{-18M^2}}{32M^3} \longrightarrow 0. \tag{2.4}$$

The correct score cost is

$$\boxed{\begin{aligned} K_{w_M^\tau} &= 2 + 4\tau\epsilon e^{-2M^2} \\ &\quad + \epsilon^2\{1 + 8M^2 + 8M^4 + (1 - 8M^2 + 8M^4)e^{-8M^2}\}, \end{aligned}} \tag{2.5}$$

and hence $K_{w_M^\tau} \sim 8\epsilon^2 M^4$.

For the lower bound on $-(q/2)\mathfrak{F}$, the $\tau = +1$ branch is the counterexample; the negative branch is useful only for auditing a two-sided or absolute estimate. Thus current-root PSD, a uniform scalar lower bound, and bounded trace/operator/HS norms do not replace K_W.

Proposition 2.2 (rank-one rotating-projector no-go)

For $\xi = (G, H) \sim N(0, \mathrm{I}_2)$ set

$$W_M^\tau = \frac{1}{2}\begin{pmatrix} 1 + \tau\cos(2MG) & \sin(2MG) \\ \sin(2MG) & 1 - \tau\cos(2MG) \end{pmatrix}. \tag{2.6}$$

Pathwise,

$$(W_M^\tau)^2 = W_M^\tau, \quad W_M^\tau \succeq 0, \quad \operatorname{rank} W_M^\tau = 1, \quad \operatorname{Tr} W_M^\tau = \|W_M^\tau\|_{\mathfrak{S}_2} = \|W_M^\tau\|_{\mathrm{op}} = 1. \tag{2.7}$$

Nevertheless,

$$\mathfrak{F}_{W_M^\tau}(h_M) = M\left[\frac{1}{2}e^{-2M^2} + \frac{\tau}{4}(1 + e^{-8M^2})\right] \sim \frac{\tau M}{4}, \tag{2.8}$$

$$K_{W_M^\tau} = 2M^4(1 + e^{-8M^2}) + 4M^2 + 2. \tag{2.9}$$

The missing datum is rotation of the eigenspace, not any eigenvalue or static matrix norm.

For a square-first covariance $W = AA^T$, write its columns as a_α. The exact product expansion is

$$\begin{aligned}\delta_\xi^2(AA^T) = \sum_\alpha \big[&(\xi \cdot a_\alpha)^2 - \|a_\alpha\|^2 - 2\xi \cdot \{(Da_\alpha)a_\alpha + a_\alpha \operatorname{div} a_\alpha\} \\ &+ 2a_\alpha \cdot D(\operatorname{div} a_\alpha) + \operatorname{Tr}[(Da_\alpha)^2] + (\operatorname{div} a_\alpha)^2\big].\end{aligned} \tag{2.10}$$

Neither PSD nor the notation AA^T gives a sign. Also $\operatorname{Tr}[(Da)^2]$ equals $\|Da\|_{\mathfrak{S}_2}^2$ only for symmetric Da.

The failure in Propositions 2.1–2.2 is registered as

`NG-2026-07-28-A13-RANDOM-W-HS-ONLY-SCORE-TRANSFER.` (2.11)

3 Trace-corrected diagonal-to-decoupled interpolation

Condition again on the strict past. Let X, X' be independent standard Gaussians in $\mathbb{R}^d$, let

$$Z_\rho = \rho X + \sqrt{1-\rho^2}\, X', \qquad 0 \le \rho \le 1, \tag{3.1}$$

and set

$$y(x,z) = b + Ax + R(z), \qquad S = AA^*. \tag{3.2}$$

Write $a_i = Ae_i$, $r_i = \partial_i R$, and

$$c(z) = \operatorname{Tr}(A^* DR(z)) = \sum_i \langle a_i, r_i(z)\rangle. \tag{3.3}$$

The family of traces is compatible precisely when an actual decoupled trace/forest δ_0 satisfies

$$\boxed{\delta_\tau = \delta_0 + 2c}, \tag{3.4}$$

and one transports

$$\delta_\rho = \delta_0 + 2\rho c. \tag{3.5}$$

The physical use of (3.4) is a theorem obligation: δ_0 may not be defined formally after deleting R-063 owners.

Define

$$\ell_\rho(x,z) = \frac{1}{2}\|y(x,z)\|^2 - \frac{1}{2}\operatorname{Tr} S - \frac{1}{2}\delta_\rho(z), \tag{3.6}$$

$$\Phi(\rho) = \log \mathbb{E}_{\gamma_\rho} e^{-q\ell_\rho}, \tag{3.7}$$

where γ_ρ is the joint Gaussian law of (X, Z_ρ). For Theorem 3.1, take $q \ge 0$, assume R and δ_0 are cylindrical C^3 functions, and assume the exponential moments of ℓ_ρ together with every derivative appearing in (3.8)–(3.15) are finite uniformly on $0 \le \rho \le 1$. The identities are first proved under these hypotheses; standard approximation may be used only when it preserves these bounds.

Theorem 3.1 (exact diagonal interpolation)

Let

$$p_\rho = A^* y, \qquad s_\rho = DR^* y - \frac{1}{2}\nabla\delta_\rho, \qquad \Gamma_\rho = p_\rho \cdot s_\rho. \tag{3.8}$$

Then

$$\boxed{\frac{d}{d\rho}\mathbb{E}_{\gamma_\rho}\ell_\rho = 0}, \tag{3.9}$$

and, for the tilted probability law

$$d\nu_\rho = \frac{e^{-q\ell_\rho}}{\mathbb{E}_{\gamma_\rho} e^{-q\ell_\rho}} d\gamma_\rho, \tag{3.10}$$

one has

$$\boxed{\Phi'(\rho) = q^2 \mathbb{E}_{\nu_\rho}\Gamma_\rho.} \tag{3.11}$$

In components,

$$\Gamma_\rho = \sum_j \langle y, a_j\rangle \left\{ \langle y, r_j\rangle - \frac{1}{2}\partial_j\delta_0 - \rho \sum_i \langle a_i, r_{ij}\rangle \right\}. \tag{3.12}$$

At the decoupled endpoint, with $K_q = (\mathrm{I} + qS)^{-1}$,

$$\boxed{\begin{aligned}\Phi(0) = \; & -\frac{1}{2}\log\det{}_2(\mathrm{I}+qS) \\ & + \log \mathbb{E}_z \exp\left\{ \frac{q}{2}\delta_0(z) - \frac{q}{2}\langle b + R(z), K_q(b + R(z))\rangle \right\}.\end{aligned}} \tag{3.13}$$

Consequently,

$$\boxed{\Phi(1) = \Phi(0) + q^2 \int_0^1 \mathbb{E}_{\nu_\rho}\Gamma_\rho \, d\rho.} \tag{3.14}$$

Proof. Price's formula gives

$$\frac{d}{d\rho}\mathbb{E}_{\gamma_\rho} f = \mathbb{E}_{\gamma_\rho} \sum_i \partial_{x_i}\partial_{z_i} f. \tag{3.15}$$

Here $\partial_\rho \ell_\rho = -c$ and $\sum_i \partial_{x_i}\partial_{z_i}\ell_\rho = c$, proving (3.9). Applying the same operator to $e^{-q\ell_\rho}$ cancels both linear trace terms and leaves $q^2\Gamma_\rho e^{-q\ell_\rho}$, proving (3.11). At $\rho = 0$, integrate the independent X square to obtain (3.13), then integrate (3.11). □

The theorem converts the diagonal problem to two explicit gates:

$$\log \mathbb{E} \exp\left\{ \frac{q}{2}\delta_0 - \frac{q}{2}\langle b + R, K_q(b + R)\rangle \right\} \le r_0, \tag{3.16}$$

$$\int_0^1 \mathbb{E}_{\nu_\rho}\Gamma_\rho \, d\rho \le G. \tag{3.17}$$

If both costs are predictable, then

$$\Phi(1) \le \frac{q^2}{4}\|S\|^2_{\mathfrak{S}_2} + r_0 + q^2 G. \tag{3.18}$$

No accepted authority proves (3.16)–(3.17) for the production cluster. In particular, (3.12) differentiates the decoupled trace and mixed trace and therefore exposes the second-/third-jet forest rather than removing it.

4 A nonlinear square-first normalizer needs a mean-debt owner

Proposition 4.1 (exact trace-compatible mean-debt no-go)

Let $G \sim N(0,1)$ and

$$b = 0, \qquad A = a\varepsilon, \qquad R(G) = \varepsilon(G^2 - 1). \tag{4.1}$$

The diagonal current and its realized tangent covariance are

$$Y(G) = \varepsilon(G^2 + aG - 1), \qquad S_{\mathrm{real}}(G) = \varepsilon^2(2G + a)^2. \tag{4.2}$$

The trace difference has the exact transport

$$\delta_\tau = 4\varepsilon^2 G^2 + 4a\varepsilon^2 G = \delta_0 + 2c, \tag{4.3}$$

where $\delta_0 = 4\varepsilon^2 G^2$ and $c = 2a\varepsilon^2 G$. Nevertheless the diagonal packet is

$$L_{\mathrm{full}} = \frac{\varepsilon^2}{2}\{(G^2 + aG - 1)^2 - (2G + a)^2\}, \tag{4.4}$$

and exact moments give

$$\boxed{\mathbb{E}L_{\mathrm{full}} = -\varepsilon^2}, \tag{4.5}$$

$$\boxed{\mathbb{E}S_{\mathrm{real}}^2 = \varepsilon^4(a^4 + 24a^2 + 48).} \tag{4.6}$$

By Jensen,

$$\log \mathbb{E}e^{-qL_{\mathrm{full}}} \geq q\varepsilon^2. \tag{4.7}$$

Hence the universal extension

$$\log \mathbb{E}e^{-qL_{\mathrm{full}}} \leq \frac{q^2}{4}\mathbb{E}S_{\mathrm{real}}^2 \tag{4.8}$$

is false whenever

$$0 < \varepsilon^2 < \frac{4}{q(a^4 + 24a^2 + 48)}. \tag{4.9}$$

At $q = 10/9$, $a = 1$, and $\varepsilon = 1/10$, the two sides obey

$$\log \mathbb{E}e^{-qL_{\mathrm{full}}} \geq \frac{1}{90}, \qquad \frac{q^2}{4}\mathbb{E}S_{\mathrm{real}}^2 = \frac{73}{32400}, \tag{4.10}$$

with exact violation margin 287/32400.

More generally, any inequality with an extra remainder r requires

$$\boxed{r \geq -q\mathbb{E}L_{\mathrm{full}} - \frac{q^2}{4}\mathbb{E}\|S_{\mathrm{real}}\|_{\mathfrak{S}_2}^2.} \tag{4.11}$$

An $O(\varepsilon^2)$ negative mean cannot be paid by an $O(\varepsilon^4)$ covariance-square cost. The production theorem must cancel this mean through its complete mean/covariance/forest packet or pay it from an explicit source/sextic owner.

This failure is registered as

$$\texttt{NG-2026-07-28-A13-UNIVERSAL-NONLINEAR-TANGENT-SQUARE-FIRST-NORMALIZER.} \tag{4.12}$$

It is an abstract complete-trace method no-go, not a production-action counterexample.

5 The smallest stationary scalar cross-output model

Work on the normalized circle with $k = 1$ and

$$X(x) = A + a\cos x + b\sin x + c\cos 2x + d\sin 2x, \tag{5.1}$$

where $(a, b) \sim N(0, v\mathrm{I}_2)$, $(c, d) \sim N(0, w\mathrm{I}_2)$, and the pairs are independent. Put

$$z = \frac{a - ib}{2}, \quad u = \frac{c - id}{2}, \quad R = |z|^2, \quad S = |u|^2, \quad C = \operatorname{Re}(z^2\bar{u}), \quad \gamma = v + 4w. \tag{5.2}$$

Assume first that $v, w > 0$. Then $2R/v$ and $2S/w$ are independent unit exponentials and $|C| \le R\sqrt{S}$. All polynomial moment identities below extend continuously to $v, w \ge 0$; the zero-variance endpoints are interpreted by that extension, not by dividing by v or w. This is a stationary scalar two-frequency model for the first cross-output obstruction. No full A1 embedding is claimed here.

For $J = X\partial_x X$, the positive output coefficients are

$$\frac{\widehat{J}(1)}{i} = Az + \bar{z}u, \quad \frac{\widehat{J}(2)}{i} = z^2 + 2Au, \quad \frac{\widehat{J}(3)}{3i} = zu, \quad \frac{\widehat{J}(4)}{2i} = u^2. \tag{5.3}$$

All conjugate outputs and their cross covariances are retained.

Let G be an independent copy of the random Fourier part and set $Y_X = X\partial_x G$. If

$$x_0 = A, \quad x_{\pm1} = z, \bar{z}, \quad x_{\pm2} = u, \bar{u}, \qquad q_{\pm1} = v/2, \quad q_{\pm2} = w/2, \tag{5.4}$$

the realized whole-output covariance is

$$\boxed{S_X(n, m) = \sum_{r=\pm1,\pm2} r^2 q_r x_{n-r}\overline{x_{m-r}}.} \tag{5.5}$$

In particular,

$$\operatorname{Tr} S_X = \gamma(A^2 + 2R + 2S). \tag{5.6}$$

Theorem 5.1 (exact complete k:$2k$ packet and square margin)

The complete covariance-normal packet

$$P = \frac{1}{2}\{\|J\|_2^2 - \operatorname{Tr} S_X\} \tag{5.7}$$

is exactly

$$\boxed{\begin{aligned} P = {} & A^2R + 4A^2S - \frac{\gamma A^2}{2} + 6AC \\ & + R^2 + 10RS + 4S^2 - \gamma(R + S). \end{aligned}} \tag{5.8}$$

It is centered. Moreover, with

$$H = \mathbb{E}\|S_X\|_{\mathfrak{S}_2}^2, \tag{5.9}$$

one has

$$\boxed{\begin{aligned} H = {} & A^4\left(\frac{v^2}{2} + 8w^2\right) \\ & + A^2(v^3 + 10v^2w + 16vw^2 + 16w^3) \\ & + \frac{5}{4}v^4 + v^3w + 25v^2w^2 + 16vw^3 + 20w^4, \end{aligned}} \tag{5.10}$$

$$\boxed{\begin{aligned}\operatorname{Var}(P) = \frac{1}{4}\{&A^4v^2 + 16A^4w^2 + 2A^2v^3 + 20A^2v^2w \\ &+ 32A^2vw^2 + 32A^2w^3 + 2v^4 + 2v^3w \\ &+ 42v^2w^2 + 32vw^3 + 32w^4\}.\end{aligned}} \tag{5.11}$$

The square-first leading margin is the positive identity

$$\boxed{H - 2\operatorname{Var}(P) = \frac{v^4 + 16v^2w^2 + 16w^4}{4}.} \tag{5.12}$$

It is strict when $v + w > 0$ and zero only for the degenerate cluster.

Proof. Fourier convolution gives (5.3), and Parseval gives (5.8). Write the four columns of the map $G \mapsto X\partial_x G$ as ℓ_r; then

$$S_X = \sum_r \ell_r \otimes \ell_r, \qquad \|S_X\|^2_{\mathfrak{S}_2} = \sum_{r,s} |\langle \ell_r, \ell_s\rangle|^2. \tag{5.13}$$

Wick expansion gives (5.10)–(5.11), and subtraction gives (5.12). The primary certificate derives the Gram matrix by exact trigonometric integration; the independent certificate reconstructs it from Fourier convolution without importing the primary. ☐

For $q \downarrow 0$ from the right,

$$\log \mathbb{E}e^{-qP} = \frac{q^2}{2}\operatorname{Var}(P) - \frac{q^3}{6}\mathbb{E}P^3 + O(q^4). \tag{5.14}$$

All coefficients in the exact polynomial $\mathbb{E}P^3$ are positive for $A^2, v, w \geq 0$. Thus (5.12) and the cubic term both favor the live target

$$\boxed{\log \mathbb{E}e^{-qP} \overset{?}{\leq} \frac{q^2}{4}H, \qquad q \geq 0.} \tag{5.15}$$

This is only a one-sided small-q conclusion. A positive exponential moment of P need not exist, so no two-sided analytic neighborhood is being claimed. Four declared finite points have positive margins at both 24×24 and 48×48 Gauss–Laguerre/Bessel orders in the primary JSON. This is only a two-order sign-stability check: the numerical margins are not claimed exact or converged, and the finite scan does not prove (5.15).

6 Sharp pointwise payment and its summability obstruction

Define the artificial positive cluster payment

$$\mathfrak{D}_{\text{pay}} = \left(\frac{9}{10}A^2 + 4w\right)R + vS. \tag{6.1}$$

It is not identified with a canonical production source or sextic owner. Let

$$Q_v(A, R) = A^2(R - v/2) + R^2 - vR. \tag{6.2}$$

Theorem 6.1 (one-payment cross-resonance bound)

Pointwise,

$$P + \mathfrak{D}_{\text{pay}} \geq Q_v(A, R) + 4Q_w(A, S). \tag{6.3}$$

Indeed, the difference is bounded below by

$$\frac{9}{10}A^2R + 10RS + 6AC \geq 10R\left(\sqrt{S} - \frac{3|A|}{10}\right)^2 \geq 0. \tag{6.4}$$

The coefficient 9/10 is sharp within this pointwise architecture.

For every $q \geq 0$, the one-pair floor and centered exponential calculation give

$$\log \mathbb{E} e^{-qQ_v(A,R)} \leq \frac{qv^2}{4} + t - \log(1+t), \qquad t = \frac{qA^2 v}{2}. \tag{6.5}$$

Since R, S are independent, (6.3)–(6.5) imply

$$\boxed{\begin{aligned} \log \mathbb{E} e^{-q(P+\mathfrak{D}_{\mathrm{pay}})} \leq q\left(\frac{v^2}{4} + w^2\right) \\ + \{t_1 - \log(1+t_1)\} + \{t_2 - \log(1+t_2)\}, \end{aligned}} \tag{6.6}$$

where $t_1 = qA^2 v/2$ and $t_2 = 2qA^2 w$. Consequently,

$$\boxed{\log \mathbb{E} e^{-q(P+\mathfrak{D}_{\mathrm{pay}})} \leq q\left(\frac{v^2}{4} + w^2\right) + q^2 A^4 \left(\frac{v^2}{8} + 2w^2\right).} \tag{6.7}$$

The theorem is exact but unusable as a termwise three-dimensional production payment. Since

$$\mathbb{E}\mathfrak{D}_{\mathrm{pay}} = \frac{9}{20} A^2 v + \frac{5}{2} vw, \tag{6.8}$$

the physical k packet contains

$$k^2 A^2 v_k \asymp A^2 \langle k \rangle^{-2}. \tag{6.9}$$

Its three-dimensional shell mass grows as 2^j. By contrast,

$$k^2 v_k^2,\ k^2 v_k w_k :\ 2^{-3j}, \qquad k^4 A^4 v_k^2 :\ 2^{-j}, \tag{6.10}$$

and the remaining displayed covariance-square terms decay still faster.

Thus the pointwise coefficient 9/10 cannot be reduced, and the resulting baseline cannot be summed over every pair, or over a positive-density subset of every shell, with bounded multiplicity. This does not exclude a sparse selection or a larger contraction-connected signed cluster. The failure is registered as

$$\texttt{NG-2026-07-28-A13-CROSS-RESONANCE-POINTWISE-BASELINE-PAYMENT}. \tag{6.11}$$

7 Proof-route evidence map

Route	Verdict	Reusable conclusion
Random-W Gaussian score	proved	The entire derivative forest is $\delta_\xi^2 W$ and costs $K_W/(128\sqrt{\eta\zeta})$.
Static PSD/trace/HS control	failed	Uniformly positive scalar and fixed rank-one projector fixtures retain an unbounded signed score.
Chaos–Sobolev payment	conditional	It is dimension free and exact in one dimension, but arbitrary admissible controls have no declared Malliavin derivative budget.
Trace-corrected diagonal interpolation	proved	The mixed trace cancels only when $\delta_\rho = \delta_0 + 2\rho\,\mathrm{Tr}(A^*DR)$; the remaining tilted cross-score contains the differentiated forest.
Universal realized-covariance square cost	failed	A negative $O(\varepsilon^2)$ mean debt cannot be paid by an $O(\varepsilon^4)$ square.
Stationary scalar $k{:}2k$ model	advanced	Exact whole-output H has strict second-order room and the third-order coefficient is favorable.
Bare all-q $k{:}2k$ normalizer	open	Local expansion and finite falsifiers pass; neither a proof nor a counterexample is known.
Pointwise cross payment	proved but retired globally	Its sharp 9/10 baseline coefficient recreates the nonsummable $\lvert k\rvert^{-2}$ loss.
Full production cluster	open	Cameron–Martin mean, trace, rational recovery, visits, R-063 forest, source, and terminal sextic remain coupled.

The append-only exploration ledger records the exact questions, findings, boundaries, and revisit conditions. This table is a researcher-facing map, not private search reasoning.

8 Devil's-advocate audit

1. **Objection: PSD of W makes the random correction favorable.**

   UPHELD AGAINST THE CLAIM. Propositions 2.1–2.2 keep all static matrix invariants fixed while the current-root eigenspace rotation produces either sign. The lower-form counterexample is the positive-score branch.

2. **Objection: the DW, D^2W terms may be estimated separately.**

   UPHELD AGAINST THE CLAIM. R-107 already shows that separated second-jet companions grow quadratically while their signed sum cancels. Theorem 1.1 packages them before estimation; (1.17) is only a sufficient cost when those full tensor norms are genuinely available.

3. **Objection: (1.5) licenses replacing the Wick forest inside an exponential normalizer.**

   DISMISSED. Theorem 1.1 is an expectation/form identity. R-109's oscillatory exponential no-go remains binding.

4. **Objection: transporting the trace proves the diagonal normalizer.**

   UPHELD AGAINST THE CLAIM. Trace transport only cancels the linear Price term. The tilted cross-score (3.12) has no sign, and Proposition 4.1 shows that a negative mean debt can dominate the covariance-square cost.

5. **Objection: $H > 2\,\mathrm{Var}(P)$ proves (5.15) for every q.**

   UPHELD AGAINST THE CLAIM. It proves only the right small-q coefficient. The finite quadrature points are falsifiers, not a theorem.

6. **Objection: Theorem 6.1 closes the physical cross mode.**

   UPHELD AGAINST THE CLAIM. Its artificial payment contains the sharp baseline term $(9/10)A^2R$, whose shell mass grows as 2^j. It is a method boundary, not a production payment.

7. **Objection: the scalar and $k{:}2k$ fixtures refute Nelson.**

   DISMISSED. They omit production-specific companions or isolate a method architecture. The complete source action and the live bare cluster route are not refuted.

8. **Objection: a Fourier/Parseval sign or conjugate-output factor could create the positive margin.**

   DISMISSED. The primary route retains the $\pm 1, \pm 2$ inputs and all conjugate outputs before exact trigonometric integration. The independent route reconstructs the current, trace, and whole-output Gram matrix from a separate Fourier-polynomial engine. Both give the same mean, variance, third moment, H, and factor in (5.12).

9. **Objection: the shell verdict mixes dimensions or homogeneity.**

   DISMISSED WITH THE DECLARED MODEL SCOPE. Equations (6.8)–(6.10) compare quantities after the same derivative/covariance scaling. The three-dimensional lattice count contributes 2^{3j}, turning the displayed $|k|^{-2}$ baseline into 2^j, while the covariance-square and square-first terms have exponents -3 and -1. No scalar-circle calculation is being promoted to a full A1 tensor identity.

10. **Objection: the finite quadrature is neither converged nor a proof.**

    VALID WITH MITIGATION. Each declared point is evaluated at orders 24 and 48, and only positivity of both margins is asserted. The values are not used as theorem constants, no convergence rate or global infimum is claimed, and the all-q route stays open.

11. **Objection: hard-coded formulas could mask a shared algebraic error.**

    DISMISSED FOR THE CLAIMED CERTIFICATES. The primary script derives the packet and moments symbolically. The standard-library independent script does not import it and instead performs Fourier convolution and Gaussian polynomial expectation. The verifier compares only independently generated outputs and separately pins every source and result hash.

12. **Objection: degenerate limits contradict the formulas.**

    DISMISSED. The fixed-W limit gives $K_W = 2\|W\|^2_{\mathfrak{S}_2}$; $q \downarrow 0$ gives the right-sided expansion (5.14); the independent samples include $A = 0$ and $w = 0$; and the exact polynomial identities extend continuously to $v = 0$ or $w = 0$. Strictness in (5.12) is claimed only when $v + w > 0$.

9 Authorities and scope firewall

The load-bearing authorities are R-063 (balanced Taylor/Wick forest and deterministic-shift reconstruction), R-104 (lossless fixed-chart owner assembly), R-105 (one-pair and physical cross-mode boundary), R-107 (coherent whole-output covariance and same-root residual), R-108 (complete cluster normal form and covariance-order audit), and R-109 (all-amplitude one-pair square-first normalizer and fixed-W score transfer).

Proved here are Theorems 1.1–1.2, Corollary 1.3, Propositions 2.1–2.2, Theorem 3.1, Proposition 4.1, Theorem 5.1, and Theorem 6.1. Not proved are the production identification of $\delta_\xi^2 W$ with a predictable no-multiplicity R-063 atom, a cutoff-uniform K_W shell ledger for arbitrary controls, the all-q bare $k{:}2k$ bound (5.15), tensorization to the full field, the adapted complete-cluster lower form, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson, removals, an interacting measure, Sector-A closure, or a higher tier.

10 Executable verification

The primary certificate uses exact symbolic differentiation, Gaussian moments, trigonometric integration, whole-output Gram reconstruction, and a declared finite Gauss–Laguerre/Bessel falsifier. The independent certificate imports neither primary code nor symbolic/scientific array libraries. It reconstructs the current and trace from Fourier convolution, evaluates Gaussian moments with a separate polynomial engine, and checks the random-W fixtures through direct Gaussian quadrature.

The manifest-pinned reproduction command, exact child/integrated contracts, and PDF QA boundary are recorded in the result footer below.

Result footer

```
Result ID: A13-CLASSII-RANDOM-W-SKOROHOD-DIAGONAL-CROSSMODE-BOUNDARY
Claim:     A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION
Ledger ID: R-110
Status:    T4 ANALYTIC/EXACT/EXECUTED ADVANCE AND METHOD BOUNDARY;
           NELSON AND SECTOR A OPEN
Scope:     Finite cutoff, fixed positive floor; conditional cylindrical
           random-W form, exact diagonal interpolation, and the smallest
           stationary scalar two-frequency k:2k output model; no full A1
           embedding of that scalar model is asserted.
Precise statement: the random-W second jet equals one double Gaussian
           divergence and has the Young-optimized K_W form cost; trace-compatible
           diagonal interpolation leaves one explicit tilted cross-score;
           the scalar k:2k model has a strict small-q square-first margin,
           while its architecture-sharp pointwise repair is nonsummable for
           positive-density bounded-multiplicity production allocation.
Dependencies: R-063, R-104, R-105, R-107, R-108, R-109.
Evidence grade: ANALYTIC/EXACT/EXECUTED; TWO NON-IMPORTING ROUTES.
Reproduction command:
           E:\Dev\TECT.venv\Scripts\python.exe ‘
           codes/foundations/a13_classii_random_w_skorohod_diagonal_crossmode_boundary_verify.py
Expected output:
           primary 40/40; independent 49/49; integrated 111/111;
           aggregate 200/200; PASS.
Negative records:
           NG-2026-07-28-A13-RANDOM-W-HS-ONLY-SCORE-TRANSFER;
           NG-2026-07-28-A13-UNIVERSAL-NONLINEAR-TANGENT-SQUARE-FIRST-NORMALIZER;
           NG-2026-07-28-A13-CROSS-RESONANCE-POINTWISE-BASELINE-PAYMENT.
Falsification gate: failure of any double-divergence, chaos multiplier,
           oscillatory/projector, interpolation, mean-debt, Fourier packet,
           whole-output covariance, moment, payment, shell, independent,
           hash, PDF, ledger, count, or scope assertion.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement:
           the bare all-q k:2k bound, full adapted production cluster,
           OVERLAP_src, Nelson, removals, measure, and Sector A remain open.
Next required action: prove or falsify the bare all-q complete k:2k
           square-first inequality, then identify its contraction-connected
           extension with the owner-preserving R-063 forest and a one-use
           source/sextic ledger without the pointwise baseline payment.
```